

CSE 209 Electrical Circuits Project

Course Outcome: CO4

Program Outcome: PO1

Cognitive Level: C3

Psychomotor Level: P2, P3

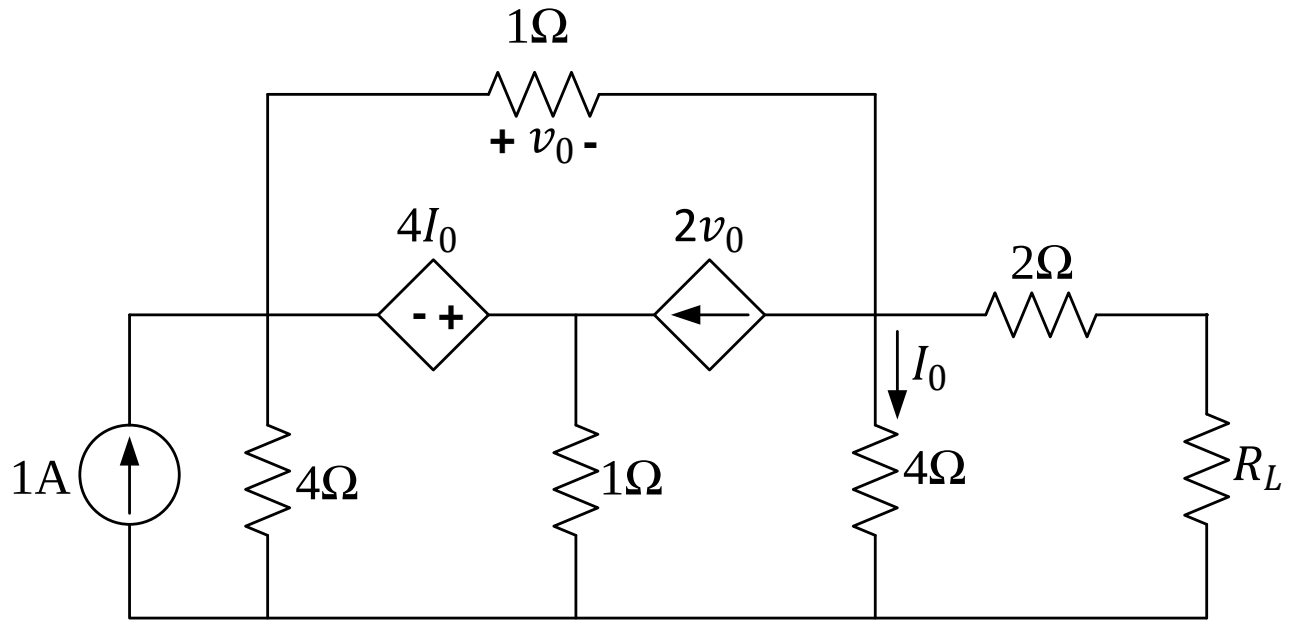
Affective Level: A2

Knowledge Profile: K1, K3

Complex Engineering Problem: EP1, EP2

PSpice Analysis for Maximum Power Transfer

1. Using PSpice Simulation, determine the Thevenin's equivalent of the circuit looking from the load resistance R_L .
2. From the Thevenin's equivalent circuit, theoretically determine the value of load resistance R_L for maximum power transfer. Using PSpice Simulation of the Thevenin's equivalent circuit with R_L for maximum power transfer, determine the value of maximum power transferred to R_L .
3. Using PSpice Simulation with resistance sweep, determine the value of R_L for maximum power transfer and the corresponding maximum power.
4. Compare the value of R_L and maximum power obtained in steps 2 and 3.



Marks Distribution

Assessment Area	Mark
C3: Cognitive: Applying	2
P2: Psychomotor: Manipulation	1
P3: Psychomotor: Precision	1
A2: Affective: Responding	1
Total	5

Project Report will include

1. Theoretical Calculation
2. PSpice Simulation (Screenshots)
3. Comparison (Table)